

✓ Proučiti stranicu

<https://poloclub.github.io/transformer-explainer/>

i odgovoriti na pitanja:

- Kako promjena temperature utječe na rezultat?
- Kako promjena samplinga utječe na rezultat?

```
from transformers import pipeline
classifier = pipeline('sentiment-analysis')
```

```
No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb0f.
Using a pipeline without specifying a model name and revision in production is not recommended.
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it as secret in y
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
config.json: 100%                               629/629 [00:00<00:00, 12.8kB/s]

model.safetensors: 100%                         268M/268M [00:02<00:00, 227MB/s]

Loading weights: 100%                           104/104 [00:00<00:00, 468.14it/s, Materializing param=pre_classifier.weight]
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster downloads.
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher ra
tokenizer_config.json: 100%                      48.0/48.0 [00:00<00:00, 2.75kB/s]

vocab.txt:    232k/? [00:00<00:00, 11.5MB/s]
```

```
classifier('The pizza is not that great but the crust is awesome.')
```

```
[{'label': 'POSITIVE', 'score': 0.9998461008071899}]
```

```
results = classifier(["We are very happy to show you the 🤗 Transformers library.",
                      "We hope you don't hate it."])
```

```
for result in results:  
    print(f"label: {result['label']], with score: {round(result['score'], 4)}")
```

```
label: POSITIVE, with score: 0.9998  
label: NEGATIVE, with score: 0.5309
```

✓ Primjer 1

preuzeti od predavača file hotel_reviews.csv

```
import pandas as pd  
from transformers import pipeline
```

```
from google.colab import drive  
drive.mount('/content/drive')
```

Mounted at /content/drive

```
path = "/content/drive/MyDrive/Colab Notebooks/hotel_reviews.csv"
```

```
hoteli = pd.read_csv (path)  
pd.set_option("display.max_colwidth", None)  
  
hoteli.head(15)
```

	text	label
0	Breakfast was tasty	positive
1	Comfortable bed	positive
2	Bad location	negative
3	The room was dirty	negative
4	Great location near the center	positive
5	Staff were very friendly	positive
6	The bed was uncomfortable	negative
7	The room was dirty	negative
8	The hotel was decent but the staff were not helpful	negative
9	Breakfast was tasty	positive
10	Staff were rude	negative
11	Comfortable bed	positive
12	The room was dirty	negative
13	Staff were very friendly	positive
14	Staff were rude	negative

```

texts = hoteli["text"].tolist()
true_labels = hoteli["label"].tolist()

# model
sentiment = pipeline("sentiment-analysis")

# predikcije
predictions = sentiment(texts)

# pretvori output
pred_labels = []

for p in predictions:
    pred_labels.append(p["label"].lower())

```

```
# ispis par primjera
for i in range(15):
    print(texts[i])
    print("TRUE:", true_labels[i])
    print("PRED:", pred_labels[i])
    print("-" * 40)
```


No model was supplied, defaulted to distilbert/distilbert-base-uncased-finetuned-sst-2-english and revision 714eb0f.
Using a pipeline without specifying a model name and revision in production is not recommended.

Loading weights: 100%

104/104 [00:00<00:00, 500.34it/s, Materializing param=pre_classifier.weight]

Breakfast was tasty

TRUE: positive

PRED: positive

Comfortable bed

TRUE: positive

PRED: positive

Bad location

TRUE: negative

PRED: negative

The room was dirty

TRUE: negative

PRED: negative

✓ Što je problem s primjenom ovog modela?

Great location near the center

Ne postoji kategorija neutral, koristite gotov model koji ga nema - ograničenje modela

PRED: positive

Ručno stvaranje datuma i vremena
Stavljamo u datim modelima tri kategorije, positive, negative, neutral

TRUE: positive

PRED: positive

<https://huggingface.co/cardiffnlp/twitter-roberta-base-sentiment>

The bed was uncomfortable

TRUE: negative

PRED: negative

✓ Primer 2

The room was dirty

```
import pandas as pd
```

```
from transformers import pipeline
```

```
the_hotel = pipeline('text-classification', model='distilbert-base-uncased-finetuned-sst-2-english')
```

```
hoteli = pd.read_csv (path)
```

```
pd.set_option("display.max_colwidth", None)
```

```
hoteli.head(15)
```

```
.....
```

Staff were rude

TRUE: negative

PRED: negative

	text	label
Comfortable bed TRUE: positive		
0	Breakfast was tasty	positive
The room was dirty TRUE: negative	Comfortable bed	positive
2	Bad location	negative
Staff were very friendly TRUE: positive	The room was dirty	negative
4	Great location near the center	positive
5	Staff were very friendly	positive
Staff were rude		
6	The bed was uncomfortable	negative
7	The room was dirty	negative
8	The hotel was decent but the staff were not helpful	negative
9	Breakfast was tasty	positive
10	Staff were rude	negative
11	Comfortable bed	positive
12	The room was dirty	negative
13	Staff were very friendly	positive
14	Staff were rude	negative

```

texts = hoteli["text"].tolist()
true_labels = hoteli["label"].tolist()

# model
sentiment = pipeline(
    "sentiment-analysis",
    model="cardiffnlp/twitter-roberta-base-sentiment"
)

# predikcije
predictions = sentiment(texts)

# pretvori output
pred_labels = []

```

```
for p in predictions:
    pred_labels.append(p["label"].lower())

# ispis par primjera
for i in range(50):
    print(texts[i])
    print("TRUE:", true_labels[i])
    print("PRED:", pred_labels[i])
    print("-" * 40)
```


Loading weights: 100%

201/201 [00:02<00:00, 103.03it/s, Materializing param=roberta.encoder.layer.11.output.dense.weight]

RobertaForSequenceClassification LOAD REPORT from: cardiffnlp/twitter-roberta-base-sentiment

Key	Status		
roberta.embeddings.position_ids	UNEXPECTED		

Notes:

- UNEXPECTED :can be ignored when loading from different task/architecture; not ok if you expect identical arch.

Breakfast was tasty

TRUE: positive

PRED: label_2

Comfortable bed

TRUE: positive

PRED: label_2

Bad location

TRUE: negative

PRED: label_0

The room was dirty

TRUE: negative

PRED: label_0

Great location near the center

TRUE: positive

PRED: label_2

Staff were very friendly

TRUE: positive

PRED: label_2

The bed was uncomfortable

TRUE: negative

PRED: label_0

The room was dirty

TRUE: negative

PRED: label_0

The hotel was decent but the staff were not helpful

TRUE: negative

PRED: label_0

Breakfast was tasty

TRUE: positive

PRED: label 2

```
-----  
Staff were rude  
TRUE: negative  
PRED: label_0  
-----  
Comfortable bed  
TRUE: positive  
PRED: label_2  
-----  
The room was dirty  
TRUE: negative  
PRED: label_0  
-----  
Staff were very friendly  
TRUE: positive  
PRED: label_2  
-----  
Staff were rude  
TRUE: negative  
PRED: label_0  
-----  
Staff were rude  
TRUE: negative  
PRED: label_0  
-----  
The room was clean and spacious  
TRUE: positive  
PRED: label_2  
-----  
Staff were very friendly  
TRUE: positive  
PRED: label_2  
-----  
Staff were very friendly  
TRUE: positive  
PRED: label_2  
-----  
Staff were very friendly  
TRUE: positive  
PRED: label_2  
-----  
The room was dirty  
TRUE: negative  
PRED: label_0  
-----  
Staff were rude  
TRUE: negative  
PRED: label_0
```

```
-----  
Bad location  
TRUE: negative  
PRED: label_0  
-----  
Breakfast was tasty  
TRUE: positive  
PRED: label_2  
-----  
The room was clean and spacious  
TRUE: positive  
PRED: label_2  
-----  
Comfortable bed  
TRUE: positive  
PRED: label_2  
-----  
Great location near the center  
TRUE: positive  
PRED: label_2  
-----  
Breakfast was tasty  
TRUE: positive  
PRED: label_2  
-----  
Comfortable bed  
TRUE: positive  
PRED: label_2  
-----  
Breakfast was terrible  
TRUE: negative  
PRED: label_0  
-----  
The room was clean and spacious  
TRUE: positive  
PRED: label_2  
-----  
Staff were rude  
TRUE: negative  
PRED: label_0  
-----  
The bed was uncomfortable  
TRUE: negative  
PRED: label_0  
-----  
Comfortable bed  
TRUE: positive  
PRED: label_2
```

```
-----  
Staff were rude  
TRUE: negative  
PRED: label_0  
-----  
Staff were very friendly  
TRUE: positive  
PRED: label_2  
-----  
Great location near the center  
TRUE: positive  
PRED: label_2  
-----  
Comfortable bed  
TRUE: positive  
PRED: label_2  
-----  
Great location near the center  
TRUE: positive  
PRED: label_2  
-----  
The room was dirty  
TRUE: negative  
PRED: label_0  
-----  
The bed was uncomfortable  
TRUE: negative  
PRED: label_0  
-----  
The room was clean and spacious  
TRUE: positive  
PRED: label_2  
-----  
Good value but far from the city  
TRUE: positive  
PRED: label_1  
-----  
Bad location  
TRUE: negative  
PRED: label_0  
-----  
Great location near the center  
TRUE: positive  
PRED: label_2  
-----  
The room was okay but not very clean  
TRUE: negative  
PRED: label_0
```

Great location near the center
 TRUE: positive
 PRED: label_2

za sljedeći multijezični dataset pronadjite gotov model koji može napraviti analizu sentimenta. Model mora imati barem tri kategorije.

Staff were rude
 TRUE: negative

```
import pandas as pd

data = [
    # ENGLISH
    ("The room was clean and spacious", "positive"),
    ("The staff were rude and unhelpful", "negative"),
    ("The hotel was okay, nothing special", "neutral"),

    # GERMAN
    ("Das Zimmer war sauber und geräumig", "positive"),
    ("Das Personal war unfreundlich", "negative"),
    ("Das Hotel war in Ordnung, nichts Besonderes", "neutral"),

    # FRENCH
    ("La chambre était propre et spacieuse", "positive"),
    ("Le personnel était impoli", "negative"),
    ("L'hôtel était correct, sans plus", "neutral"),
]

df = pd.DataFrame(data, columns=["text", "label"])

texts = df["text"].tolist()
labels = df["label"].tolist()
```

```
df.head(10)
```

	text	label
0	The room was clean and spacious	positive
1	The staff were rude and unhelpful	negative
2	The hotel was okay, nothing special	neutral
3	Das Zimmer war sauber und geräumig	positive
4	Das Personal war unfreundlich	negative
5	Das Hotel war in Ordnung, nichts Besonderes	neutral
6	La chambre était propre et spacieuse	positive
7	Le personnel était impoli	negative
8	L'hôtel était correct, sans plus	neutral

✓ Bert multijezični model

<https://huggingface.co/nlptown/bert-base-multilingual-uncased-sentiment>

```
from transformers import pipeline

classifier = pipeline(
    "sentiment-analysis",
    model="nlptown/bert-base-multilingual-uncased-sentiment"
)

results = classifier(texts)

# ispis
for i in range(len(texts)):
    stars = results[i]["label"]
    confidence = results[i]["score"]

    print("TEXT:", texts[i])
    print("PRED:", stars)
    print("CONF:", round(confidence, 3))
    print("-" * 50)
```

Loading weights: 100%

201/201 [00:00<00:00, 527.10it/s, Materializing param=classifier.weight]

TEXT: The room was clean and spacious

PRED: 4 stars

CONF: 0.432

TEXT: The staff were rude and unhelpful

PRED: 1 star

CONF: 0.52

TEXT: The hotel was okay, nothing special

PRED: 3 stars

CONF: 0.754

TEXT: Das Zimmer war sauber und geräumig

PRED: 5 stars

CONF: 0.462

TEXT: Das Personal war unfreundlich

PRED: 1 star

CONF: 0.457

TEXT: Das Hotel war in Ordnung, nichts Besonderes

PRED: 3 stars

CONF: 0.619

TEXT: La chambre était propre et spacieuse

PRED: 5 stars

CONF: 0.411

TEXT: Le personnel était impoli

PRED: 2 stars

CONF: 0.372

TEXT: L'hôtel était correct, sans plus

PRED: 3 stars

CONF: 0.681

✓ Primjer 3

```
import pandas as pd
import gradio as gr
from transformers import pipeline
```

```
data = [
    "The room was very clean and tidy",
    "The bathroom was dirty and smelled bad",
    "Staff were very friendly and helpful",
    "The receptionist was rude and unhelpful",
    "The hotel is in a great location near the center",
    "The hotel is far from everything",
    "Breakfast was delicious and fresh",
    "The food was cold and tasteless",
    "Great location but the room was dirty",
    "Nice staff but terrible breakfast",
]
```

```
df = pd.DataFrame(data, columns=["text"])
df.head()
```

	text
0	The room was very clean and tidy
1	The bathroom was dirty and smelled bad
2	Staff were very friendly and helpful
3	The receptionist was rude and unhelpful
4	The hotel is in a great location near the center

```
import gradio as gr
from transformers import pipeline
```

```
# Initialize the zero-shot classification pipeline
classifier = pipeline("zero-shot-classification", model="facebook/bart-large-mnli")
```

```
# Define the classification function
```

```
def classify_text(text, labels):
    labels = labels.split(",")
    result = classifier(text, candidate_labels=labels)
    return {label: score for label, score in zip(result["labels"], result["scores"])}
```

```
# Create the Gradio interface
interface = gr.Interface(
    fn=classify_text,
    inputs=[
        gr.Textbox(lines=2, placeholder="Enter text here..."),
        gr.Textbox(lines=1, placeholder="Enter comma-separated labels here...")
    ],
    outputs=gr.Label(num_top_classes=3),
    title="Zero-Shot Text Classification",
    description="Classify text into labels without training data.",
)

# Launch the interface
interface.launch()
```